

Module 06: Learning in Robotics

Learning Objectives

1. Define **Learning** within the context of Robotics.
2. Analyze how Learning interacts with other components of the Active Inference framework.
3. Apply specific constraints of Robotics to the formal definition of Learning.

Introduction

This module explores **Learning**. In the **Robotics** curriculum, we approach this topic with a focus on specific applications and theoretical depth appropriate for the audience. Learning is a critical component of the 8-part Active Inference spine, bridging the gap between Action and Communication.

Key Concepts

1. Learning as a Markov Blanket Boundary

How does Learning define the boundary between the agent and the environment?

2. Generative Models of Learning

What parameters involved in Learning must be optimized to minimize variational free energy?

3. Active Inference Dynamics

How does the process of Learning drive the perception-action loop?

Applications

In Robotics, we see Learning manifest in: * **Specific Example 1:** A self-driving vehicle's prediction module learns pedestrian behavior models from millions of miles of driving data,

updating the parameters of its trajectory-prediction neural network to minimize prediction error on held-out scenarios; over time, the model learns context-dependent priors -- pedestrians near crosswalks are likely to cross, pedestrians at bus stops are likely to remain stationary -- that reduce the free energy of the generative model by encoding structured prior knowledge that pure physics-based prediction cannot capture. * **Specific Example 2:** An autonomous drone performing package delivery in a new city learns a wind disturbance model specific to the local urban canyon environment by recording control corrections applied during initial flights; a neural network maps GPS position and altitude to expected wind vector, and after 50 delivery flights the learned model reduces position tracking error by 40% compared to the nominal controller, demonstrating how autonomous agents update their generative models from operational experience to improve future performance without manual recalibration.

Conclusion

Understanding Learning allows us to better model complex adaptive systems. In the next module, we will expand on this foundation.